

a) Označme $(\vec{v}_1, \vec{v}_2, \vec{v}_3) = (1, x, x^2)$.

Spočítáme G-S ortogonalizaci ortonormální bázi $(\vec{u}_1, \vec{u}_2, \vec{u}_3)$.

$$u_1 = \frac{1}{\sqrt{\int_1^5 1^2 dx}} = \frac{1}{2}$$

$$v_2 - w_2 = x - \frac{1}{2} \int_1^5 \frac{x}{2} = x - 3$$

$$u_2 = \frac{x-3}{\|x-3\|} = \frac{x-3}{\sqrt{\int_1^5 (x-3)^2 dx}} = \frac{\sqrt{3}}{4} (x-3)$$

$$\begin{aligned} v_3 - w_3 &= x^2 - \frac{1}{2} \int_1^5 \frac{x^2}{2} dx - \frac{\sqrt{3}}{4} (x-3) \int_1^5 x^2 \cdot \frac{\sqrt{3}}{4} (x-3) dx = \\ &= x^2 - 6(x-3) - \frac{31}{3} = x^2 - 6x + \frac{23}{3} \end{aligned}$$

$$u_3 = \frac{x^2 - 6x + \frac{23}{3}}{\sqrt{\int_1^5 (x^2 - 6x + \frac{23}{3})^2 dx}} = \frac{\sqrt{5}}{16} (3x^2 - 18x + 23)$$

$$(\vec{u}_1, \vec{u}_2, \vec{u}_3) = \left(\frac{1}{2}, \frac{\sqrt{3}}{4} (x-3), \frac{\sqrt{5}}{16} (3x^2 - 18x + 23) \right)$$

$$= \left(\frac{1}{2}, 0,433x - 1,299, 0,419x^2 - 2,516x + 3,214 \right)$$

↓
0,5

b) Spočítame přes vzorec: (Tvrz. 8.54)

$$\begin{aligned} W &= \left\langle \frac{1}{2}; \sin x \right\rangle \frac{1}{2} + \left\langle 0,433x - 1,299; \sin x \right\rangle (0,433x - 1,299) + \\ &+ \left\langle 0,419x^2 - 2,516x + 3,214; \sin x \right\rangle (0,419x^2 - 2,516x + 3,214) \\ &= 0,128 \cdot 0,5 + (0,433x - 1,299)(-1,493) + \\ &+ (-0,725)(0,419x^2 - 2,516x + 3,214) = \\ &= 0,064 - 0,646x + 1,939 - 0,052x^2 + 0,315x \\ &- 0,402 = \underline{-0,052x^2 - 0,331x + 1,601} \end{aligned}$$

$$\begin{aligned} c) \quad &\| \sin x + 0,052x^2 + 0,331x - 1,601 \| = \\ &= \sqrt{\int_1^5 (\sin x + 0,052x^2 + 0,331x - 1,601)^2 dx} = 0,319 \end{aligned}$$

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